

Interview-Prep : MySQL – 50 Most Frequently Asked Questions and 10 Query Based Questions

Part 1: The Questions

Beginner (Basic Concepts & CRUD)

1. What is MySQL and what are its key features?
2. What is the default port for MySQL?
3. Explain the difference between SQL and MySQL.
4. How do you see all the databases available on the MySQL server?
5. What command is used to select a database?
6. How do you view the structure of a table?
7. What is a Primary Key?
8. What is a Foreign Key?
9. Explain the CHAR vs. VARCHAR data types.
10. What is the difference between INT and BIGINT?
11. How do you create a table in MySQL?
12. What is an Alias in MySQL?
13. How do you filter unique values in a query?
14. What does the LIKE operator do?
15. What is the purpose of the LIMIT clause?

Intermediate (Joins, Logic & Constraints)

16. Name and explain the four types of Joins in MySQL.
17. What is the difference between WHERE and HAVING?
18. What is a UNIQUE constraint?
19. What are Aggregate Functions? List five examples.
20. What is a Subquery?
21. What is the difference between DELETE, TRUNCATE, and DROP?
22. How do you handle NULL values in MySQL?

23. What is an Index, and why is it used?
24. Explain the difference between a Clustered and Non-Clustered index.
25. What is a View?
26. What is a Trigger?
27. What is a Stored Procedure?
28. Explain the difference between UNION and UNION ALL.
29. What is a Composite Key?
30. What is the difference between the = and LIKE operators?
31. How do you find the second highest salary in an employees table?
32. What is the IN operator?
33. How do you rename a table or a column?
34. What is the use of the BETWEEN operator?
35. How do you concatenate two strings in MySQL?

● **Advanced (Architecture, Performance & Administration)**

36. What are ACID properties?
37. Explain the difference between InnoDB and MyISAM storage engines.
38. What is Database Normalization and why is it used?
39. What is Denormalization?
40. How do you optimize a slow query in MySQL?
41. What is the EXPLAIN statement used for?
42. What are the different levels of Normalization (1NF, 2NF, 3NF)?
43. What is a Deadlock and how does MySQL handle it?
44. Explain MySQL Master-Slave replication.
45. What is a Savepoint?
46. What are "Dirty Reads"?
47. How do you export and import data using mysqldump?
48. What is the performance_schema?
49. What is the difference between FLOAT and DECIMAL?

50. What is a Common Table Expression (CTE)?

Part 2: The Query Challenges

1. The Unsold Products: List all products (ID and Name) that have never been sold. (Requires a LEFT JOIN or NOT EXISTS).
2. Manager Names: In an employees table with id, name, and manager_id, write a query to show each employee's name and their manager's name.
3. Department Top Earners: Find the name and salary of the highest-paid employee in each department.
4. Active Customers: Find customers who placed an order in both January 2024 and February 2024.
5. Running Total: Given a sales table (date, amount), calculate the cumulative (running) total of sales ordered by date.
6. Find the Second Highest: Find the second-highest salary from an employees table without using LIMIT.
7. Overlapping Salaries: Find employees who earn more than the average salary of their own department.
8. The "At Least N" Join: List departments that have more than 5 employees.
9. Product Sales Mix: List all products and the total quantity sold, including products with zero sales (ensure zero is shown instead of NULL).
10. Duplicate Detection: Identify all email addresses that appear more than once in a users table.